Sliding Gate Installation – Pre-Installation Checklist & Detailed Installation Process

1. Purpose

This procedure provides a comprehensive checklist and step-by-step installation process for installing a sliding gate (manual or automatic). It ensures safe installation, proper alignment, smooth operation, and long-term reliability.

PART A – PRE-INSTALLATION CHECKLIST

A. Document Verification

Item	Status
Approved shop drawings available	□
Structural drawings approved	□
Manufacturer's installation manual available	□
Material Inspection Report (MIR) approved	□
Site measurements verified	□
Inspection and Test Plan (ITP) available	□
Risk Assessment and Method Statement approved	□

B. Material Inspection

Inspect all materials before installation.

Gate Leaf

- □ Dimensions as per approved drawing
- □ Surface finish free from scratches
- □ No bending or twisting
- □ Proper welding quality
- □ Paint/powder coating intact

Track

- □ Straight
- □ Rust free
- □ Correct length
- □ Proper section

Rollers

- □ Bearings rotate freely
- □ No damage
- □ Correct size
- □ Greased

Guide Rollers

- □ Rubber/nylon intact
- □ Bearings smooth
- □ Mounting holes correct

Accessories

- □ End stopper
- □ Gate catcher
- □ Guide brackets
- □ Mounting plates
- □ Fasteners
- □ Anchor bolts

C. Civil Work Verification

Foundation

Verify:

- □ Foundation dimensions
- □ Concrete grade
- □ Concrete fully cured (minimum 7–28 days as specified)
- □ Surface level
- □ No cracks
- □ Anchor bolts positioned correctly

Typical Foundation

- Width: As per drawing
- Depth: According to structural design
- Reinforcement installed
- Level tolerance:
    - ±3 mm over 3 m

D. Site Readiness

Verify

- □ Area cleaned
- □ No construction debris
- □ Safe access available
- □ Lighting available
- □ Drainage completed
- □ Utilities identified

E. Dimensional Verification

Measure and confirm:

Item	Required
Gate opening width	Drawing
Gate height	Drawing
Run-back distance	Gate width + counterbalance (cantilever) or full travel distance (track-mounted)
Track straightness	±2 mm
Track level	±2 mm
Plumpness of posts	±2 mm/m

F. Equipment Availability

- □ Laser level
- □ Spirit level
- □ Measuring tape
- □ Total station (if required)
- □ Hammer drill
- □ Core drill
- □ Welding machine
- □ Grinder
- □ Torque wrench
- □ Impact wrench
- □ Chain block
- □ Crane/Forklift (if required)

G. Safety Checklist

- □ Work permit issued
- □ Toolbox talk completed
- □ Barricading installed
- □ PPE available
- □ Fire extinguisher available
- □ Lifting equipment certified
- □ Lifting plan approved

PART B – INSTALLATION PROCESS

Step 1 – Site Inspection

Activities

- Verify approved drawings.
- Check gate opening dimensions.
- Inspect foundation.
- Confirm installation reference lines.
- Verify concrete strength.

- Confirm clear travel path.

Inspection Point

Foundation must comply with approved dimensions.

Step 2 – Marking

Using:

- Laser level
- Chalk line
- Total station

Mark:

- Track centreline
- Roller locations
- Guide bracket positions
- Motor base location
- End stopper position

Tolerance

±2 mm

Step 3 – Install Track (Track-Mounted Gates)

Procedure

1. Clean foundation.
2. Position track.
3. Align using laser.
4. Check level.
5. Check straightness.
6. Weld or bolt track.
7. Recheck alignment.

Acceptance Criteria

- Track level
- Straight
- Securely fixed

Step 4 – Install Roller Assemblies

Procedure

1. Position roller assemblies.
2. Align with track.
3. Tighten anchor bolts.
4. Verify spacing.
5. Grease bearings.

Inspection

Rollers rotate freely.

Step 5 – Lift the Gate

Equipment

- Crane
- Forklift
- Chain block

Procedure

1. Attach lifting slings.
2. Lift slowly.
3. Protect painted surface.
4. Position over rollers.
5. Lower carefully.

Safety

Never stand beneath suspended loads.

Step 6 – Position the Gate

Check

- Vertical alignment
- Horizontal alignment
- Roller engagement
- Track contact

Operate manually.

Gate should move smoothly.

Step 7 – Install Guide Rollers

Guide rollers should:

- Prevent lateral movement
- Allow free travel
- Not create excessive friction

Adjust until:

- No wobbling
- Smooth operation

Step 8 – Install End Stopper

Purpose

Prevent gate from running off the track.

Procedure

- Position at travel limit.
- Secure by welding or bolting.
- Verify full contact.

Step 9 – Install Gate Catcher

The catcher:

- Supports the gate when closed.
- Reduces roller load.
- Prevents vibration.

Adjust until:

- Gate seats fully.
- No excessive force required.

Step 10 – Install Motor (Automatic Gate)

Install

- Motor base
- Gear motor
- Drive pinion

Verify

- Motor alignment
- Shaft level
- Fastener torque

Step 11 – Install Rack

Procedure

1. Open gate halfway.
2. Fix first rack section.
3. Maintain correct gear mesh.
4. Install remaining sections.
5. Check continuity.
6. Verify full travel.

Acceptance

No tight spots.

Step 12 – Electrical Installation

Install:

- Power cable
- Control panel
- Limit switches
- Safety photocells
- Flashing beacon
- Emergency stop
- Earth connection

Testing

- Voltage
- Continuity
- Insulation resistance
- Grounding

Step 13 – Adjustment

Adjust:

- Roller height
- Guide rollers
- Motor position
- Rack clearance
- Limit switches

Gate should:

- Move smoothly
- Stop accurately
- Not bind
- Not vibrate

Step 14 – Functional Testing

Perform at least 10 complete opening and closing cycles.

Verify:

Test	Acceptance
Smooth movement	Pass
Noise level	Minimal
Motor current	Within manufacturer's limit
Safety photocell operation	Functional
Emergency stop	Functional
Manual release	Functional
Gate closing	Complete
Gate opening	Complete

Step 15 – Final Inspection

Check:

- All bolts tightened
- Welds complete
- Paint damage repaired
- Lubrication completed
- Labels installed
- Safety devices operational
- Area cleaned

PART C – QUALITY CHECKPOINTS

Activity	Inspection
Foundation	Level and dimensions
Track	Straightness and level
Rollers	Free rotation
Gate alignment	Plumb and level
Guide rollers	Proper clearance
Rack	Correct gear engagement
Motor	Alignment
Electrical	Functional tests
Final operation	Smooth travel

PART D – SAFETY PRECAUTIONS

- Wear appropriate PPE (helmet, gloves, safety shoes, eye protection, and high-visibility clothing).
- Use only certified lifting equipment and inspect slings before use.
- Keep personnel clear of suspended loads during lifting.
- Barricade the work area and display warning signs.
- Lock out and tag out electrical supplies before wiring.

- Do not energize the motor until all mechanical components are securely installed.
- Use proper lifting techniques to avoid manual handling injuries.
- Follow the manufacturer's installation and maintenance instructions at all times.

PART E – FINAL ACCEPTANCE CRITERIA

The sliding gate installation is considered complete when:

- The gate is installed according to approved drawings and manufacturer specifications.
- The gate travels smoothly through its full opening and closing range without binding or excessive noise.
- Track alignment, level, and structural supports are within specified tolerances.
- All fasteners are tightened to the required torque and protective coatings are intact.
- Safety devices (photocells, limit switches, emergency stop, manual release, etc.) are installed and function correctly.
- Electrical wiring has been tested and complies with applicable codes.
- Functional testing has been successfully completed and documented.
- The installation area is clean, safe, and ready for handover to the client.

Sign-Off

Item	Status (✓/X)	Remarks
Site Ready		
Foundation Approved		
Electrical Ready		
Gate Components Complete		
Safety Devices Ready		
Installation Approved		